04 - Research Methodology

How the experiments are designed

Knowledge-pack rule: every statement below is derived from files in this project or is explicitly labelled as planned, missing or not verified. Nothing here is an invented citation, dataset, result or feature.

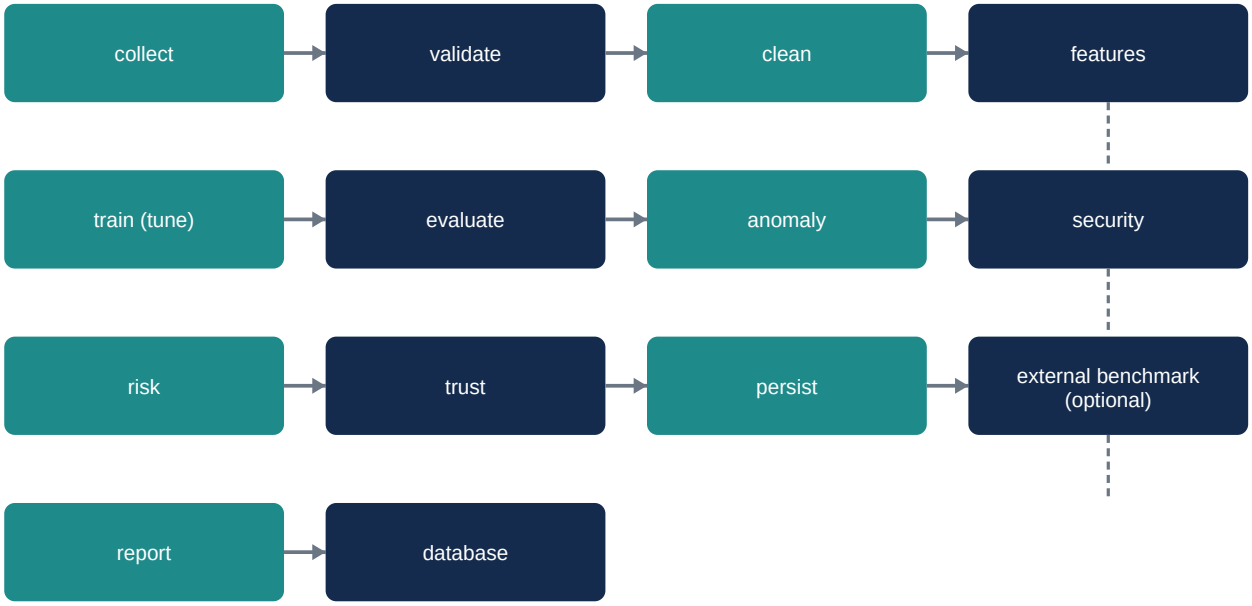
Source files used for this document

documentation/methodology.md; src/pipeline.py; config/settings.yaml; report/chapters/chapter_3_methodology.md

Research design

Design-science style computer-science experiment plus a quantitative survey component. Everything is seeded and logged; each model's metadata stores a data hash, seed and library versions.

Pipeline stages (in order)



Stages as implemented in src/pipeline.py; each is timed and exported as a Prometheus metric.

Experimental protocol

Item	Setting in config/settings.yaml	Why
Seed	42	Reproducibility
Split	chronological train 0.7 / validation 0.15 / rest test	Prevents leakage from the future
Tuning	RandomizedSearchCV, n_iter=8, cv_folds=3, scoring=average_precision	Random search is efficient; average precision suits rare positives
Imbalance	class_weight='balanced' (LR), 'balanced_subsample' (RF), scale_pos_weight (XGBoost)	No resampling, so probabilities stay meaningful for risk tiers
Threshold	F1-optimal on the validation set	Test set untouched until final evaluation
Anomaly contamination	0.02	Expected outlier share; threshold taken from validation scores

Evaluation metrics

Metric	Meaning in a fraud context
Accuracy	Share of correct predictions - misleading: predicting 'legit' always gives ~98% when fraud is ~2%
Precision	Of transactions flagged as fraud, how many really are fraud
Recall	Of real frauds, how many were caught
F1	Harmonic mean of precision and recall
ROC-AUC	Ranking quality over all thresholds; can look optimistic under imbalance
PR-AUC (average precision)	Ranking quality focused on the rare positive class; used to choose the champion
False-positive rate	Share of legitimate transactions wrongly flagged = customer friction
Confusion matrix	TP, FP, FN, TN counts
Bootstrap 95% CI	Uncertainty of champion ROC-AUC (200 resamples)
McNemar test	Are two models' errors different on the same test cases

Consumer trust methodology

- 12 Likert items (1-5) in five dimensions; q12 is reverse-coded (6 - x).
- Dimension score = (mean - 1) / 4 x 100; Consumer Trust Score (CTS) = weighted mean of dimensions.
- Weights from settings.yaml: {'perceived_security': 0.3, 'privacy_assurance': 0.2, 'transparency': 0.15, 'reliability': 0.2, 'behavioural_intent': 0.15}
- Segments: Low < 50, High >= 75, otherwise Moderate.
- Reliability: Cronbach's alpha. Drivers: Spearman correlation, OLS on standardised predictors, Kruskal-Wallis, Mann-Whitney.
- Platform Security Index (PSI) = 100 x mean(recall, 1 - min(10 x FPR, 1), attack block rate, validation pass rate).
- Consumer Trust Index (CTI) = a x mean CTS + (1 - a) x PSI with a = 0.6. This is a proposed composite, not a validated instrument.

Validity threats (from the project's methodology document)

- Internal: synthetic labels reflect generator assumptions.
- External: real fraud adapts; distribution shift is not modelled.
- Construct: 12 items are a coarse proxy for trust.
- Statistical: few positives in the test set make recall estimates wide.